

BUREAU OF INDIAN STANDARDS

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भारतीय मानक मसौदा

**पशु चिकित्सा सर्जिकल उपकरण — लोहा तांबे की सुइयों से पायरो-
पंकचर — विशिष्टि**

(IS 12246 का पहला पुनरीक्षण)

Draft Indian Standard

**Veterinary Surgical Instruments — Iron Pyro-puncture
with Copper Needles — Specification**

(First Revision of IS 12246)

ICS 11.220

Veterinary Hospital Planning and Surgical
Instruments Sectional Committee, MHD 13

Last date for comments: **22 September 2026**

FOREWORD

(Formal clauses will be added later)

This Indian Standard was first published in 1988 as ‘Specification for Irons, pyro - Puncture with copper needles’. The first revision of this standard has been brought out to align the cross references to latest standards.

For the purpose of deciding whether a particular requirement of this standard is complied with the final value, observed or calculated, expressing the result of a test or analysis shall be rounded off in accordance with IS 2 : 2022 ‘Rules for rounding off numerical values (second revision)’. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

1 SCOPE

This Indian standard covers the requirements for irons, pyro-puncture with copper needles for veterinary use.

2 REFERENCES

The standards given below contain provisions which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of these standards.

IS No.	Title
IS 2102 (Part 1) : 1993	General Tolerances Part 1 Tolerances for Linear and Angular Dimensions Without Individual Tolerance Indications (<i>third revision</i>)
IS 1068-2025	Metallic and Other Inorganic Coatings — Electrodeposited Coatings of Nickel, Nickel Plus Chromium, Copper Plus Nickel and of Copper Plus Nickel Plus Chromium (Fourth Revision)
IS 2500 (Part 1) : 2000	Sampling procedures for inspection by attributes: Part 1 sampling schemes indexed by acceptance quality limit (AQL) for lot - By - Lot inspection (<i>third revision</i>)

3 MATERIAL

a) Head, rod and spring	Carbon steel having 0.45 to 0.55% carbon
b) Spring holder and pin	Mild steel
c) Ferrule and washer	Brass having minimum 60% copper (brass 60/40)
d) Insulator	Suitable timber
e) Needle	Copper

4 SHAPE AND DIMENSIONS

4.1 Shall Conform to Fig. 1.

4.2 The tolerance on dimensions of working ends shall conform to ‘fine’ class of deviation and the tolerance for remaining dimensions shall conform to ‘medium’ class of deviation of IS 2102 (Part 1).

5 MANUFACTURE WORKMANSHIP AND FINISH

5.1 The edges shall be even, rounded and nowhere sharp. The spring holder shall provide a close fit to the spring. The ferrule shall be firmly fixed to the handle. The riveting and welding shall be sound. The spring shall be accurately and evenly hardened and tempered. The spring shall function satisfactorily.

The pin shall fit into its corresponding part. The needle shall be pointed and shall fit into its corresponding part of the iron.

5.2 The surface of the working ends shall be free from all defects like burrs, pits, cracks, nicks and waviness when examined under 10 X magnification. All surfaces shall be free from pores, crevices and grinding marks.

5.3 The instrument shall be supplied free from residual scales, acid, grease and grinding and polishing materials. Compliance with these requirements shall be checked under 2 X magnification.

6 HARDNESS

The head, rod and spring shall be hardened and tempered to give a hardness of 450 to 550 *HV*.

7 REQUIREMENTS

The steel parts shall be plated nickel over copper conforming to service condition number 2 of IS 1068. The brass parts shall be nickel plated conforming to service condition number 2 of IS 1068.

8 TESTS

8.1 Spring Test

The distance between the free end of the spring and the nearest point on the head shall be measured. The spring shall be pressed to the head and released 12 times. On completion of the test, the measured distance shall not have altered.

8.2 Load Test

The load required to be applied to the free end of the spring to bring it in contact with the lower face of the head shall be 680 g.

8.3 Corrosion Test

8.3.1 The instrument shall be cleaned with a suitable organic solvent such as carbon tetrachloride, in order to remove any trace of grease before carrying out the test.

8.3.2 The instrument shall be boiled for 60 minutes in 5 percent aqueous solution of sodium carbonate. At the end of this period, the plating shall not show any sign of peeling off nor shall there be any rusting over the plated portion.

8.3.3 The instrument shall be placed on a perforated platform on a desiccator above a solution of sodium chloride (20 g NaCl in 100 cc of water of specific gravity of 1.15). Air shall be blown into the solution so that a dense fog is produced in the desiccator without spray being blown directly against the instrument. There shall be no sign of corrosion of the plating after treating for 24 hours.

9 SURFACE FINISH

The surface finish shall be mirror finish. The instrument should be adequately ground to remove all surface imperfections and polished to remove grinding marks, resulting in a mirror finish. The mirror finish should be effected by an appropriate procedure, such as polishing, brushing, electropolishing and mirror buffing.

10 SAMPLING

10.1 Unless otherwise agreed to between the purchaser and the supplier, the sampling procedure given in IS 2500 (Part 1) shall be followed.

10.2 Lot

All the irons of the same size, shape and belonging to the same batch of manufacture shall be grouped together to constitute a lot.

10.3 For dimensions and requirements given in **4**, **5** and **9**, a single sampling plan with General inspection level II and AQL 4 percent given in Tables 1 and 2-A of IS 2500 (Part 1) shall be followed.

10.4 For the requirement given in **7**, a single sampling plan with General inspection level I and AQL 1 percent given in Tables 1 and 2-A of IS 2500 (Part 1) shall be followed.

10.5 For the tests given in **6** and **8**, a single sampling plan with special inspection level S-3 and AQL 1 percent given in Tables 1 and 2-A of IS 2500 (Part 1) shall be followed.

11 MARKING

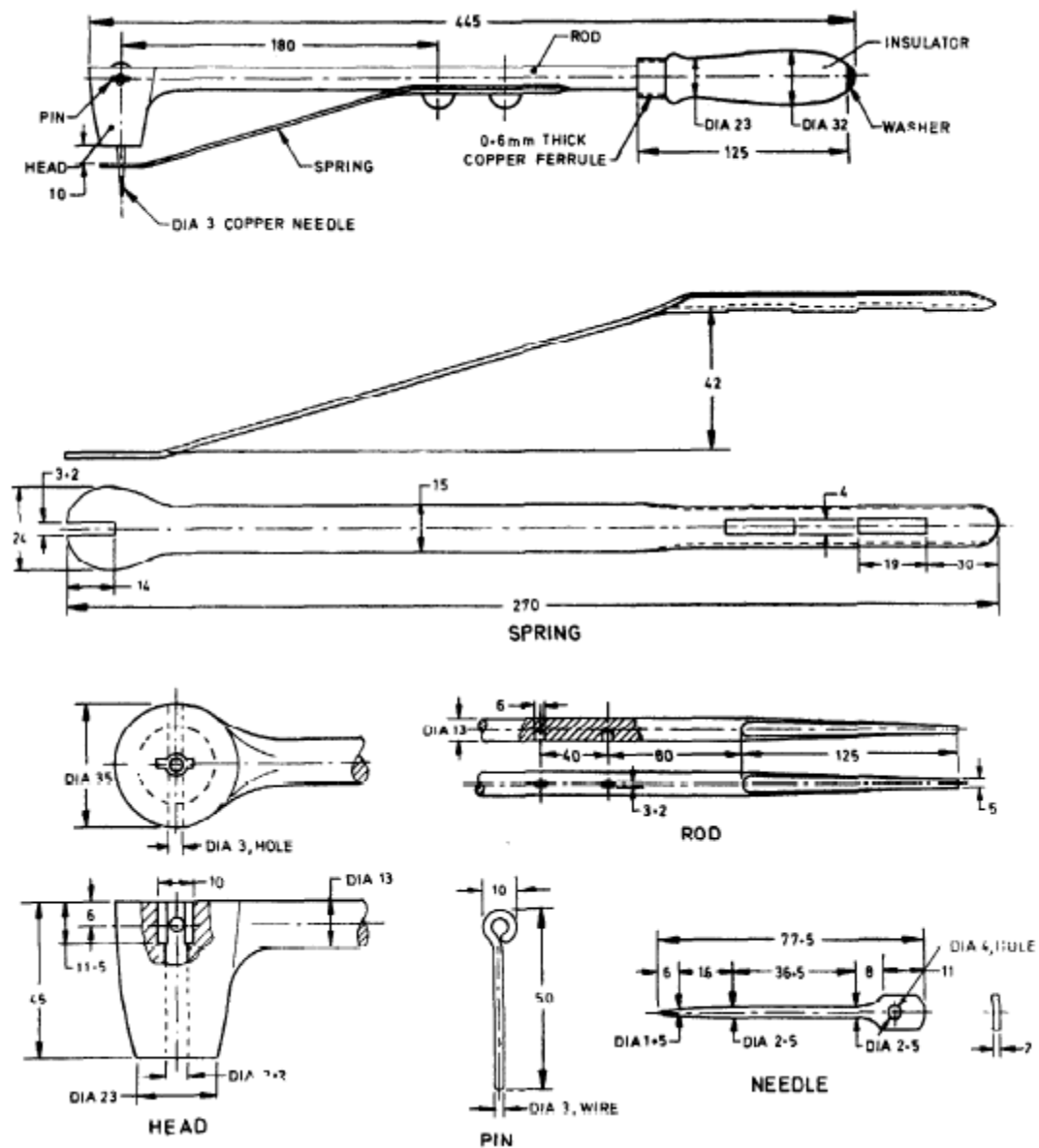
11.1 Each iron shall be legibly and indelibly marked with the manufacturer's name, initials or recognized trade-mark.

11.2 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act, 2016* and the Rules and Regulations framed thereunder, and the product(s) may be marked with the Standard Mark.

12 PACKING

A set of 6 needles shall be wrapped in Kraft paper. The empty spaces, if any, shall be filled with suitable cushioning material to prevent movement.



All dimensions in millimetres.

FIG. 1 IRON, PYROPUNCTURE WITH COPPER NEEDLE